Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ______________
Topic: Systems of Linear Inequalities by Graphing **Date:** ____________

Quiz (Carolina Reaper)

Q1. A shipping company has x and y routes. The inequality $2x + 3y \leq 12$ represents the time constraint. Which inequality represents the cost constraint if each x route costs $2 and each y route costs $3?
(A) $2x + 3y \geq 6$
(B) $2x + 3y \leq 6$
(C) $2x + 2y \leq 12$
(D) $3x + 2y \leq 12$
(E) $x + 2y \leq 6$

Q2. The system of inequalities $x + 2y \leq 8$ and $3x - 2y \geq -4$ has which of the following graphs?
(A) overlapping lines
(B) parallel lines
(C) intersecting lines
(D) perpendicular lines
(E) skew lines

Q3. What is the feasible region for the system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$?
(A) the entire xy-plane
(B) the first quadrant
(C) the second quadrant
(D) a triangle with vertices $(0, 0)$, $(0, 6)$, and $(6, 0)$
(E) a triangle with vertices $(0, 0)$, $(0, 6)$, and $(-6, 0)$

Q4. For a delivery company, x represents the number of packages delivered by truck and y represents the number of packages delivered by plane. The inequality $x \geq 10$ represents which of the following?
(A) at least 10 packages are delivered by plane
(B) at least 10 packages are delivered by truck
(C) exactly 10 packages are delivered
(D) at most 10 packages are delivered
(E) no packages are delivered

Q5. A company has two warehouses. Warehouse A has x boxes and Warehouse B has y boxes. The company can store a maximum of 50 boxes in total. Which inequality represents this situation?
(A) $x + y \leq 50$
(B) $x + y \geq 50$
(C) $x - y \leq 50$
(D) $x - y \geq 50$
(E) $2x + 2y \leq 100$

Q6. Given the system of inequalities $y \leq 2x + 3$ and $y \geq x - 2$, what is the point of intersection?
(A) $(-1, 1)$
(B) $(-1, -1)$
(C) $(1, 1)$
(D) $(-5, -7)$
(E) $(2, 2)$

Q7. A traffic flow is modeled by the inequalities $x + y \leq 15$ and $x - y \geq -5$. What is the maximum value of x?
(A) 10
(B) 15
(C) 20
(D) 5
(E) 8

Q8. For the system $2x + y \geq 4$ and $x - 2y \leq -2$, which of the following points is in the solution region?
(A) $(1, 2)$
(B) $(-1, 2)$
(C) $(1, -2)$
(D) $(-1, -2)$
(E) $(0, 0)$

Open Ended Questions (FRQ)

Q9. A transportation company needs to transport x boxes by land and y boxes by sea. The company has the following constraints: land transport costs $2 per box and sea transport costs $3 per box, with a total budget of $120. Additionally, the company can transport a maxim

Q10. A delivery company has two routes: Route A and Route B. Route A has a time constraint of $2x+3y \leq 12$ hours and a cost constraint of $3x + 2y \leq 15$ dollars, where x is the number of packages delivered by Route A and y is the number of packages delivered by Route B. Graph the system of inequalities and find the maximum number of packages that can be delivered within the given time and cost constraints.

Chef's Tips

- Use graph paper to help visualize the systems of inequalities.
- Make sure to label the axes and include a title for each graph.
- Check for any special cases, such as parallel or coincident lines.